Triangle Angles Maze ANSWER KEY

Start in the top-left room. Solve the problem, find the door with your answer, and go through it. Show your work on your answer grid.

START

1 Angles of a △. Find x.
50°, 60°, x°

$x = 120$

Angles of a △. Find x.
42°, 73°, x°

$x = 5$

4 Equilateral △. Find x.
One angle
$(3x + 15)$°

x = 15

5 Find the exterior angle.
Remote interior ∠s
45°, 65°

x = 70

$x = 250$

$x = 115$

x = 30

$x = 25$

$x = 55$

110°

2 Find the other acute angle.
Right triangle
one acute ∠: 34°

56°

3 Angles of a △. Find x.
x°, $2x$°, $3x$°

$x = 48$

Exterior ∠ theorem. Find x.
exterior ∠: 132°
remote ∠s: 57°, x°

44°

6 Isosceles △. Find a base ∠.
Vertex angle
44°

146°

$x = 45$

$x = 60$

$x = 36$

12°

136°

68°

Isosceles △. Find vertex ∠.
Base angle
37°

40°

9 △ angles. Find the largest.
$(x + 10)$°, $(2x - 20)$°, $(3x + 10)$°

80°

8 Isosceles △. Find vertex ∠.
Base angles
$(5x - 10)$°, $(3x + 14)$°

x = 32

7 Exterior ∠: $(4x - 12)$°. Find x.
Remote interior ∠s
$(2x + 8)$°, $(x + 12)$°

143°

30°

60°

100°

50°

$x = 8$

$x = \frac{172}{7}$

Right △. Find x.
Acute angles
$(3x + 6)$°, $(2x - 1)$°

$x = 35$

Equilateral △. Find x.
One angle
$(5x - 20)$°

$x = 10$

10 Isosceles △. Find x.
vertex ∠: $2x$°
base ∠s: $(x + 10)$°

x = 40

FINISH
You made it!

Triangle Angles Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **$x = 70$**	$50 + 60 + x = 180 \rightarrow x = 70$
2	Answer: **56°**	$180 - 90 - 34 \rightarrow = 56°$
3	Answer: **$x = 30$**	$6x = 180 \rightarrow x = 30$
4	Answer: **$x = 15$**	$3x + 15 = 60 \rightarrow 3x = 45 \rightarrow x = 15$
5	Answer: **110°**	$45 + 65 = 110$
6	Answer: **68°**	$180 - 44 = 136 \rightarrow \frac{136}{2} = 68°$
7	Answer: **$x = 32$**	$2x + 8 + x + 12 = 4x - 12 \rightarrow 3x + 20 = 4x - 12 \rightarrow x = 32$
8	Answer: **80°**	$5x - 10 = 3x + 14 \rightarrow x = 12 \rightarrow 180 - 2(50) = 80$
9	Answer: **100°**	$6x = 180 \rightarrow x = 30 \rightarrow 3(30) + 10 = 100$
10	Answer: **$x = 40$**	$2x + 2(x + 10) = 180 \rightarrow 4x + 20 = 180 \rightarrow x = 40$